Estimand Framework

A (Very) Brief Introduction

Gogtay NJ, Ranganathan P, Aggarwal R. (2021). Understanding Estimands. Perspect Clin Res

Executive Summary

- The estimand framework is a systematic description of the treatment effect to be quantified to answer the trial's research objective
- It has been called 'old wine in new bottles' because it is just (more) thoughtful, modified intention to treat considerations
- For most experienced trialists, who are used to defining treatment effects and sensitivity analyses, any *new* thinking typically relates to the handling of intercurrent events

Estimands

- Addendum to the ICH E9 in 2017
  - Reflects the current thinking of the FDA
- Estimand
  - A precise definition of the treatment effect being estimated
- Estimator
  - How the treatment effect will be calculated (estimated)
- Estimate
  - The resulting value used to interpret the treatment effect

Intercurrent Events

- Post-randomization event (or experience) that impacts the measurement of treatment effect
- Causes each individual to experience their own ‘path’ through the trial
  - Creates difficulty in interpretation

- Examples:
  - Finishes study exactly as protocolized
  - Starts alternative treatment (or palliative treatment)
  - Develops a side-effect that requires monitoring or intervention
  - Discontinues treatment
  - Death (for studies where death is not an outcome)
  - Rescue therapy used
  - Undergoes additional surgery/intubation

Four Elements of the Estimand

- The population
  - Precise definition of eligibility including dynamic changes to eligibility
- The variable(s)
  - Precise definition of outcome and timing of measurement(s)
- Intercurrent events
  - Events (experiences) that occur after intervention (randomization?)
- Population-level summary
  - Precise definition of the effect estimate, its form, and method

Strategies for Intercurrent Events

- Treatment Policing Strategy
  - Value of the outcome regardless of intercurrent events
  - ITT
- Composite Strategy
  - Intercurrent event woven into the outcome
  - "Proportion with delirium according to … and not reintubated or experiencing additional surgery…"
- Hypothetical Strategy
  - Assumes that the intercurrent event did not happen (needs a statistical model)
  - "Proportion who experienced or would have experienced delirium had they been able to test…"
- Principal Stratum Strategy
  - Considers measurements in subgroup(s) of the target population (by design)
  - Dynamic randomization (by former intercurrent event)
- While on Treatment Strategy
  - Evaluate treatment effect prior to the occurrence of the intercurrent event
  - Measurements stop after intercurrent event

Sensitivity Analyses

- Conducted as typically done in regards to considering the impact of different assumptions on the estimate (or even variations in the estimator)

- Could be done to consider variations in the estimand, by varying the strategy used to consider intercurrent events (or variations in the population, etc.)